YEAR 9 MATHS · CHAPTER 9 · MIND MAP

Probability and statistics

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1 Review of probability

$P(\text{event}) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$, from 0 to 1.

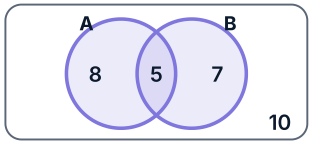
One die: $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

$P(\text{not } A) = 1 - P(A)$: $P(\text{not } 6) = \frac{5}{6}$

2 Venn diagrams and two-way tables

Fill in the overlap first.

30 students: $P(A) = \frac{13}{30}$, $P(A \cap B) = \frac{1}{6}$



3 Set notation

$A \cap B$: A and B. $A \cup B$: A or B (or both). A': not A.

From the Venn: $n(A \cup B) = 8 + 5 + 7 = 20$

$n(A') = 7 + 10 = 17$

4 Arrays for two-step experiments

List every pair of outcomes in a table.

Two dice: $6 \times 6 = 36$ outcomes

Sum of 7: 6 ways, so $P = \frac{6}{36} = \frac{1}{6}$

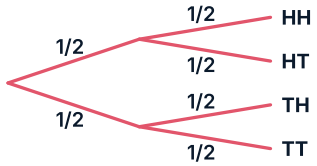
5 Tree diagrams

Multiply along the branches. Add the results you want.

Two coins: $P(\text{HH}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$P(\text{at least one head}) = 1 - \frac{1}{4} = \frac{3}{4}$

3 red and 2 blue, take 2 without replacement: $P(\text{both red}) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$



Watch out: without replacement, the second branch has one fewer.

6 Relative frequency

$\text{Relative frequency} = \dfrac{\text{times it happened}}{\text{number of trials}}$

A pin lands point-up 36 times in 120 throws: $\frac{36}{120} = 0.3$

Expected in 500 throws: $0.3 \times 500 = 150$

More trials give a better estimate.

7 Data and sampling

A sample should represent the whole population. Random samples avoid bias.

Categorical: colours, types. Numerical: discrete (counts) or continuous (measurements)

Asking only gym members how often people exercise is biased

8 Mean, median and mode

Mean: add and divide. Median: middle value in order. Mode: most common.

3, 5, 5, 8, 9: mean $\frac{30}{5} = 6$, median 5, mode 5

2, 4, 7, 9: median $\frac{4+7}{2} = 5.5$

9 Stem-and-leaf plots

The stem is the tens, the leaf is the units. Leaves go in order.

2 | 1 4 7 and 3 | 0 2 (key 2 | 4 = 24)

Data 21, 24, 27, 30, 32: median **27**

10 Grouping data into classes

Group numerical data into equal class intervals, then draw a histogram.

Classes 0–9, 10–19, 20–29 with a frequency for each

Class centre of 10–19: **14.5**

With grouped data, use the class centres to estimate the mean.

11 Range and interquartile range

Range = max − min. IQR $= Q_3 - Q_1$, the spread of the middle half.

2, 4, 5, 7, 8, 10, 12, 15: range $15 - 2 = 13$

$Q_1 = 4.5$, median 7.5, $Q_3 = 11$, so IQR $= 6.5$

12 Box plots

Draw the five-number summary: min, Q_1, median, Q_3, max. The box holds the middle 50%.

The data from card 11:

